

HEVC Video Adversarial Samples Detection via Joint Features of Compression and Pixel Domains

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Abstract—Deep learning models are currently under significant threat from adversarial attacks, while adversarial detection represents an effective means of countering such assaults. However, existing adversarial detection techniques are deficient in localizing video adversarial frames, leading to poor performance on sparse video adversarial attacks. This paper presents an approach for detecting adversarial perturbations in videos based on fusion features derived from the video compression and RGB domain. Our research begins by examining how the introduction of extensive non-natural noise during video adversarial attacks severely disrupts the spatial structure of individual frames and the motion information between frames. This disruption culminates in unnatural variations in the Coding Tree Units (CTU) partitioning during the HEVC video encoding process. Then meticulously mapping the positions and partitioning information of coding units (CU), predictive units (PU), and transformation units (TU) onto specific values and sizes, constituting the video's Compression Domain Units (CDU) features. Finally, a dual-path network utilizing both the video's CDU features and the decoded frames RGB features is employed for detecting video adversarial samples. Extensive experiments are conducted to verify the performance. The results show that the proposed scheme outperforms or rivals the state-of-the-art methods in video adversarial detection.

Index Terms—Video adversarial detection, compression feature, adaptive feature fusion, dual-path network.

I. INTRODUCTION

ARTIFICIAL intelligence systems rooted in deep neural networks have achieved remarkable successes across numerous domains in recent years. Concurrently, the emergence of adversarial examples has engendered concerns over the security of these AI systems, thereby constraining their more profound penetration into safety-critical applications. By introducing imperceptible perturbations into natural inputs, well-performing deep neural networks cannot produce correct output [1], [2], [3].

Video content understanding models are also vulnerable to adversarial attacks, causing unreliable deployment in safety-critical tasks such as dangerous behavior recognition, anomaly

detection, and illegal content screening. Compared to static image adversarial examples, numerous constituent frames within a video sample allow for categorizing video adversarial attacks into global and sparse attacks, depending on whether all frames or only a subset thereof are subjected to perturbation. Global adversarial attacks entail the application of adversarial perturbations to every frame within a given sample, exemplified by techniques such as Anti Coding Attack [4], Temporal Translation Attack [5], and Cross Model Attack [6]. This approach confers the advantage of simplicity and heightened attack success rates but reduces the inconspicuousness of perturbations. In contrast, sparse adversarial attacks involve the selective targeting of specific frames within a sample, with adversarial perturbations applied exclusively to these chosen frames, as seen in methods like AstFocus [7], CLVA [8], and Sparse Attack [9]. This strategy enhances the stealthiness of the perturbations, improves query efficiency in black-box scenarios, and reduces the complexity of optimization.

Adversarial detection methods have been proven effective in resisting adversarial attacks, where the defender discriminates between subtly perturbed adversarial samples and clean ones by mapping them into an additional feature space. For instance, the ADDNP [10] approach utilizes an NRP [11] network to reconstruct input samples. It employs a target model to compute the feature similarity between the reconstructed and original samples to differentiate them. Inspired by cryptography, the PseudoRandom Classifier [12] method transforms the original one-hot labels into multi-dimensional ones. It employs multiple models for classification voting on these transformed labels, thus detecting adversarial samples. The VASA technique [13] applies a style transfer model to samples and assesses the similarity of prediction outcomes before and after style transformation for adversarial detection purposes. A commonality among these methods lies in amplifying minuscule adversarial perturbations through mapping, thereby constructing detectors for adversarial samples.

While the previous adversarial detection methods demonstrate admirable performance in the context of image adversarial samples, they struggle to address the challenges posed by video adversarial detection, primarily due to the following three reasons:

1. Ineffective utilization of temporal information in video samples: These adversarial detection methods are designed for static images and thus inherently cannot capture temporal adversarial artifacts in videos.

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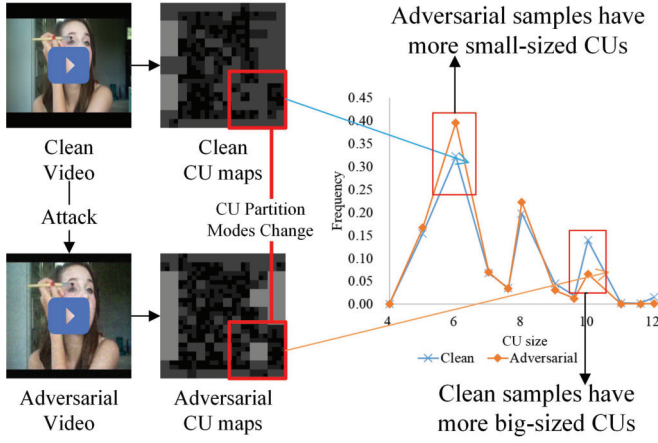


Fig. 1. The addition of adversarial perturbation can lead to changes in the partitioning mode of CU during HEVC encoding, increasing the proportion of small-sized CUs in adversarial samples, resulting in a decrease in the matching degree between semantic information and CU partitioning information.

2. The computational intensity and inferior reconstruction in transformation-based approaches: Given the significantly larger data volume associated with video samples than static images, detection methods relying on reconstruction and transformation struggle to converge effectively, leading to poor detection results.

3. The challenge of sparse attacks: Sparse attacks only introduce perturbations to selected video frames. When conducting detection on a video-level basis, the presence of a substantial number of clean frames significantly impairs the detector's performance, as it must navigate through both adversarial and non-adversarial content.

To address the issues above, this paper proposes a method for adversarial video frame detection based on the fusion of Compression Domain Units (CDU) features and Pixel Domain RGB frame features. The schematic diagram of our proposed method is shown in Fig. 1. Our intuition stems from the fact that during video encoding, the encoder jointly optimizes encoding parameters based on both spatial texture information and temporal motion information within the video, such as the selection of quantization tables, quadtree partitioning of Coding Tree Units (CTUs), and choice of intra-/inter-frame prediction modes. In the case of natural images, a high degree of consistency exists between encoding parameters and semantic image information. For example, smaller block partitions are selected in areas of complex textures, while more detailed prediction mode partitions are employed in regions with rapid motion changes. However, for adversarial samples, the added adversarial noise disrupts both inter-frame motion correlations and intra-frame semantic correlations within the video sample. Consequently, the adversarial noise influences the selection of encoding parameters, leading to a breakdown in the consistency between these parameters and the RGB semantic information.

Detecting adversarial traces becomes possible by assessing the degree of match between CDU features and RGB frame features. Moreover, given the abundance of encoding domain parameter features available for extraction within each frame, this approach enables the localization of adversarial frames,

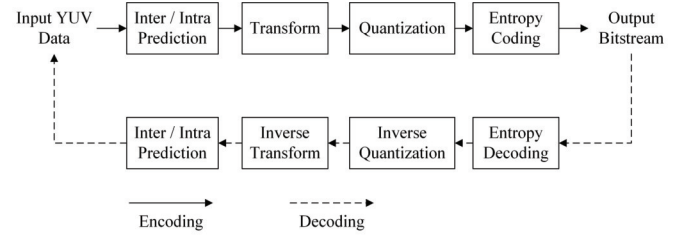


Fig. 2. HEVC coding/decoding process.

thereby significantly enhancing the detection capability against sparse video attacks.

In summary, the contributions of this paper are as follows:

1. By analyzing the differences in HEVC encoding parameters generated by adversarial samples versus clean samples within the compression domain, demonstrating that adversarial attacks significantly impact parameter selection during video encoding.

2. By meticulously mapping the values and sizes of CDU features of HEVC encoding: coding units (CU), predictive units (PU), and transformation units (TU) within the compression domain, and subsequently examining the degree of matching between CDU features and RGB features, a universal method for detecting video adversarial samples is proposed in this work.

3. To the best of our knowledge, this study constitutes the first investigation to employ compression domain features for detecting video adversarial samples. Extensive experiments reveal that our approach achieves state-of-the-art detection performance in video adversarial detection, particularly in the case of sparse adversarial video attacks.

II. RELATED WORK

During transmission and storage, video samples undergo intricate encoding and decoding processes, which yield a wealth of extractable features. This paper will focus on today's most widely used video coding format, the HEVC/H.265 standard. The first part will briefly overview the encoding and decoding processes in HEVC. The second part will concisely introduce some applications of compression domain features.

A. HEVC Encoding / Decoding Process

HEVC encoding adheres to a classic hybrid coding paradigm composed of four main stages: inter- and intra-frame prediction, transformation, quantization, and entropy coding. Fig. 2 presents a schematic representation of the encoding and decoding workflow in HEVC, where solid lines denote the forward video encoding process. In contrast, dashed lines indicate the reverse video decoding process.

The video encoding and decoding process is a lossy compression procedure. Under the constraint of a given bitrate R_{max} , the optimal encoding parameter m' is determined by iterating through the set of encoding parameters m to minimize distortion D . This optimization process can be implemented using the Lagrange optimization algorithm, as depicted by Eq.(1).

$$m' = \underset{m' \in m}{\operatorname{argmin}} (D(m') + \lambda \cdot R(m')) \quad (1)$$

Here, $D(m)$ represents the distortion under the encoding parameter setting m , while $R(m)$ denotes the bitrate under the same parameter configuration, $R(m) \leq R_{max}$. λ is the Lagrange multiplier employed during rate-distortion optimization, serving as a hyper-parameter determined based on encoding experience and statistical data. It is preset within the encoder.

Encoding in HEVC operates on the fundamental unit of Coding Tree Units (CTU). Each frame is initially partitioned into non-overlapping Coding Units (CU) via quadtree partitioning. During inter- and intra-frame prediction, Prediction Units (PU) are derived from CU, with their partitioning guided by the frame's internal texture information and inter-frame motion data, offering eight distinct partitioning patterns. Transform Units (TU) are partitioned from the CU root using a quadtree structure for transformation and quantization. Smaller CDUs are employed in energy-dense regions to preserve fine-grained texture details, while larger CDUs are used in energy-sparse areas to reduce bitrate efficiently. Under circumstances devoid of extraneous noise disturbances, CU, PU and TU's partitioning should exhibit a high degree of correspondence with information in the RGB domain. In this paper, our primary focus lies in the CDUs partitioning information during the encoding and decoding process to extract features that can facilitate the detection of adversarial traces.

B. Compression Domain Feature Application

The compression domain of videos contains a wealth of information, such as motion information, residual data, and quantization tables. Leveraging these elements for video-related tasks can result in computational savings and enable work under partial decoding conditions. For instance, employing motion vectors and residual information from the compressed domain for video action recognition [14], [15] yields accuracy comparable to RGB-based approaches but with significantly faster computation times. Utilizing encoding-domain features for video double compression detection [16], [17] is another crucial area of research. Detecting whether a video has undergone double compression can effectively ascertain if the original video has been tampered with by attackers.

III. THEORETICAL ANALYSIS

In this section, an initial analysis is conducted to explore the impact of adversarial perturbations on the encoding process. Subsequently, the mapping and modeling of compression domain features are achieved based on the variations in encoding parameters.

A. Influence of Adversarial Perturbation on Video Encoding

The lossy video encoding and decoding process can be conceptually simplified as transmission through a communication channel, as illustrated in Fig. 3. Here, the original video source ensemble is denoted by X , and upon encoding, it yields the corresponding codeword Y . Given that the encoding process involves operations like quantization and Rounding, which can be analogously regarded as introducing extra noise into the

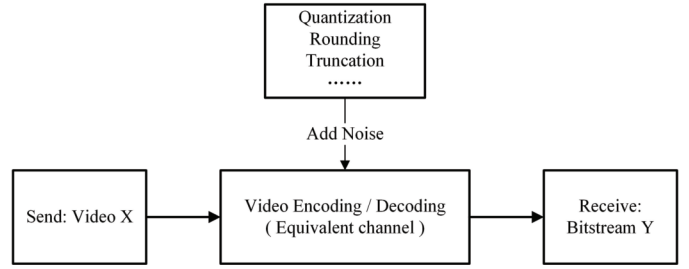


Fig. 3. Lossy video encoding and decoding process can be conceptually simplified as a communication channel.

channel, the received codeword fails to establish a one-to-one correspondence with the transmitted source. Therefore, the video encoding process can be equivalent to a lossy channel. In this channel, the required bitrate for transmitting the video is denoted by R :

$$R = I(X; Y) = I(X) - I(X|Y) \quad (2)$$

where $I(X; Y)$ represents the mutual information between X and Y , $I(X)$ denotes the entropy of the source video X , and $I(X|Y)$ signifies the conditional source entropy of X given Y , indicating the amount of information lost during encoding, this can be regarded as the distortion $D = I(X|Y)$ incurred during the encoding process.

Now, consider the scenario where adversarial noise δ is added to the original video X . For the sake of clarity, let us assume that this adversarial noise is uniformly distributed, independent, and identically distributed (i.i.d.) concerning the original video X (Indeed, extensive data analysis reveals that the covariance coefficient between the two falls below 0.001, suggesting they can essentially be considered independent), following a distribution denoted by $\delta \sim U(-\epsilon, \epsilon)$, where ϵ is the upper bound of the δ . If the same encoding and decoding parameters used for clean samples are employed, the required bitrate for the channel would then become R' :

$$\begin{aligned} R' &= I(X + \delta; Y) \\ &= I(X + \delta) - I(X + \delta|Y) \\ &= I(X) - I(X|Y) + I(\delta) - I(\delta|Y) \\ &= R + I(\delta) - I(\delta|Y) \geq R \end{aligned} \quad (3)$$

Moreover, the distortion is D' :

$$D' = I(X|Y) + I(\delta|Y) = D + I(\delta|Y) \geq D \quad (4)$$

According to the definition of conditional information entropy, $I(\delta|Y)$ is greater than 0 and smaller than $I(\delta)$, so it can be known that R' is greater than R and D' is greater than D , implying that under identical encoding parameters, the bitrate required for encoding adversarial samples would be higher, and correspondingly, the losses incurred during encoding would also be more significant.

Consequently, to ensure both an acceptable level of distortion D' and bitrate R' for adversarial video samples, it is necessary to alter the encoding method by introducing a new channel mapping symbol Y' , thereby maintaining both the distortion and bitrate within tolerable bounds.

Thus, by capturing the disparities between Y and Y' , effective detection of video adversarial traces can be achieved.

B. Influence of Adversarial Perturbation on CDU Partitioning in HEVC Encoding

In HEVC video coding schemes, the encoding problem can be abstracted as a Rate Distortion Optimization (RDO) task. Given a maximum allowed bit rate R_{max} , the goal is to minimize the distortion D . Assuming independence among the coding units subject to optimization, let B_k denote the encoding parameters (such as quantization step size, coding mode, motion vectors, etc.) for the k_{th} unit, $R_k(B_k)$ represent the corresponding encoded bits, and $D_k(B_k)$ denote the encoding distortion. Under these conditions, the RDO problem specifically entails finding a set of encoding parameters $\hat{B}$ under the constraint of R_{max} , using the Lagrange multiplier method to minimize the overall distortion:

$$\hat{B} = \underset{\{B_1, B_2, \dots, B_n\}}{\operatorname{argmin}} \left(\sum_{k=1}^n D_k(B_k) + \lambda \sum_{k=1}^n R_k(B_k) \right) \quad (5)$$

For the HEVC video coding format, the primary encoding workflow is divided into four main segments: inter- and intra-frame prediction, transformation, quantization, and entropy coding. Given that the entropy coding stage does not introduce any additional loss, it will be disregarded in the following discussion. Considering an original video X , with HEVC encoding operating on basic units called CU blocks, let x_i denote the i_{th} CU block. During inter- and intra-frame prediction, the corresponding reference pixels are denoted by RP_i , yielding the predicted residue $PR_i = x_i - RP_i$. Subsequently, the residue transforms T , such as a Discrete Cosine Transform (DCT), followed by quantization using a quantization table QT_i , yielding the quantization coefficients $QC_i = [PR_i / QT_i]$, where $[\cdot]$ denotes rounding operation.

During subsequent video decoding, the decoded block $\hat{x}_i$ is:

$$\hat{x}_i = LF [T^{-1} (QC_i \times QT_i) + RP_i] \quad (6)$$

where LF denotes in-loop filtering, T^{-1} denotes inverse transformation. At last, the distortion of this block is $D_i = \hat{x}_i - x_i$. Rounding errors, in-loop filtering effects, and integer-to-float conversions occurring during DCT transformations in the encoding and decoding process can be negligible contributors to loss.

1) *Adversarial Influence in PU Partition Pattern*: Within each smallest coding unit (CU) in HEVC, prediction unit (PU) partitioning is performed. The division of inter- and intra-prediction modes is realized by selecting different PU partitioning modes, thereby optimizing the choice of predictive pixels.

During the optimization of Eq. (5), a bitrate allocation is performed for the blocks within the CU partitioning based on their division according to inter- and intra-prediction modes [18], as exemplified by Eq. (7).

$$R_i = \frac{R_{frame} - Bit_H - Coded_{frame}}{\sum_{AllNotCodedCUs} \omega_{CU}} \times \omega_{x_i} \quad (7)$$

Here, R_i represents the allocated coding length for the x_i , R_{frame} denotes the coding length assigned to the frame containing the block x_i , Bit_H is the byte length of header files required during encoding, calculated as the average byte length of header files in already encoded blocks, and $Coded_{frame}$ is the cumulative byte length of encoded data in the frame containing the block. Lastly, ω_{CU} is a parameter related to inter- and intra-prediction, which can be calculated by the estimated Mean Absolute Difference (MAD_{x_i}). N_i denotes the number of pixels in coding block x_i .

$$MAD_{x_i} = \frac{1}{N_i} |PR_i| \quad (8)$$

The calculation of ω_{x_i} is shown in Eq. (9).

$$\omega_{x_i} = MAD_{x_i}^2 \quad (9)$$

For an adversarial CU block $x'_i = x_i + \delta_i$, its $MAD_{x'_i}$ during inter- and intra-prediction will differ due to the introduction of noise, effectively introducing an additional prediction error $e = \delta_i - \delta_{RP_i}$ where δ_{RP_i} denotes the adversarial perturbations added on reference pixels. e follows a triangular distribution with a minimum value of -2ϵ , a maximum value of 2ϵ , and a median of 0. Without loss of generality, it is assumed that the codeword length allocated to each frame during the encoding process remains unchanged, and the changes in prediction modes in other CU blocks are ignored. Hence, the coding length allocated for this specific block is denoted by R'_i . The difference DR_i between R_i and R'_i is shown in Eq. (10).

$$\begin{aligned} DR_i &= |R_i - R'_i| \\ &= |C \times (\omega_{x_i} - \omega_{x'_i})| \\ &= |C \times (MAD_{x_i}^2 - MAD_{x'_i}^2)| \\ &= \left| C \times \left(\frac{1}{N_i^2} (|PR_i|^2 - |PR_i + e|^2) \right) \right| \\ &\geq \frac{C}{N_i^2} e^2 \end{aligned} \quad (10)$$

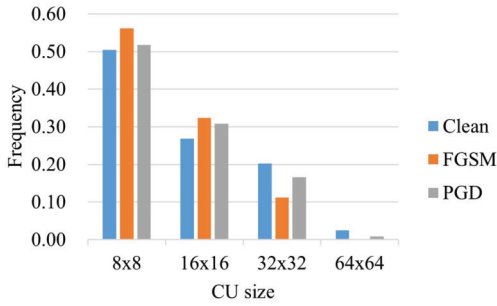
For simplicity, let C represent the constant term in the equation.

$$C = \frac{R_{frame} - Bit_H - Coded_{frame}}{\sum_{AllNotCodedCUs} \omega_{CU}} \quad (11)$$

From above analysis, after applying an adversarial perturbation with a maximum amplitude of ϵ , the difference introduced in the bitrate allocation within a CU based on inter- and intra-frame prediction is no less than $\frac{C}{N_i^2} e^2$, with an expected value of $\frac{2C}{3N_i^2} \epsilon^2$. The consequence of this is that the PU partitioning patterns in the adversarial sample will undergo noticeable changes, rendering them discernible features for detecting video adversarial samples.

2) *Adversarial Influence in TU Partition Pattern*: A crucial reference metric during TU partitioning is the quantization loss experienced by the TU block after transformation and quantization. The quantization parameter QP , which determines which quantization table to apply to the current TU block during quantization, is given by the following equation [19]:

$$QP = c_1 \times \ln(\lambda) + c_2 \quad (12)$$



(a) Histogram of CU Frequency Distribution



(b) CU maps

Fig. 4. Histograms depicting the frequency distributions of CU blocks generated by different samples in HEVC encoding.

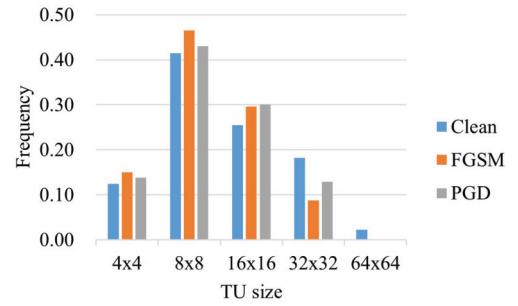
where c_1 and c_2 are set to 4.2005 and 13.7122, respectively, and λ is the lagrangian multiplier in Eq. (5).

During the adversarial attack process, the generated adversarial perturbation primarily consists of high-frequency noise, leading to an increase in the high-frequency components within the adversarial video sample. As a result, when encoding such a sample under the same QP constraint as the clean sample, adopting the identical TU partitioning would cause an increase in distortion. Consequently, during the encoding of the adversarial sample, the TU partitioning scheme tends to favor a higher proportion of smaller TU blocks, thereby refining the encoding of the adversarial noise and striving to minimize distortion as much as possible.

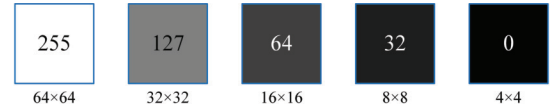
C. Generation of CDU Maps

Through the above analysis, it can deduce that during video encoding of adversarial video samples, the adversarial noise introduced influences the CDU partitioning pattern in the encoding process, resulting in variations in both the bitrate and distortion produced by the final encoded output. Therefore, this characteristic can be exploited to detect adversarial video samples.

1) *CU Map*: In the process of video encoding with HEVC, the original video frames are initially partitioned into 64×64 blocks, referred to as coding units (CU). These CUs are further refined based on the frames' specific texture and semantic features, a process known as quadtree partitioning. The CU can be subdivided into 32×32 , 16×16 , and 8×8 blocks. Employing this method, larger-sized CUs are suitable for regions with smoother textures, enhancing encoding efficiency, while smaller-sized CUs are tailored to areas with intricate textures, effectively reducing distortion. Based on the preceding analysis, the proportion of smaller-sized CUs in adversarial samples under the same semantic information context should be higher than those in clean samples due to the increased high-frequency noise resulting from the adversarial attack process.



(a) Histogram of TU Frequency Distribution



(b) TU maps

Fig. 5. Histograms depicting the frequency distributions of TU blocks generated by different samples in HEVC encoding.

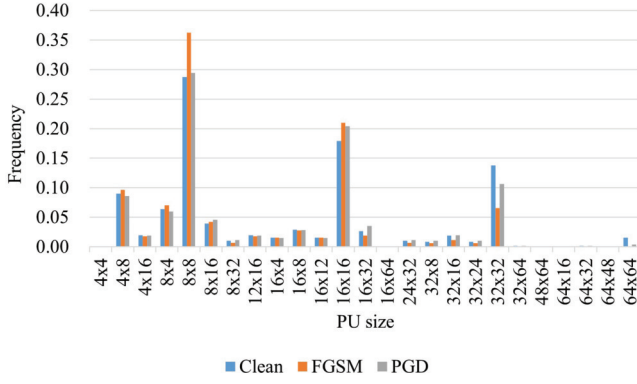
100 clean samples are selected and targeted the I3D network [20], subjecting them to FGSM [21] and PGD [22] attack methods. We then statistically analyzed the video coding characteristics of the adversarial samples. As seen in Fig. 4a, video adversarial samples generated by FGSM and PGD attacks exhibit a higher proportion of 8×8 and 16×16 small blocks compared to clean samples while showing a lower share of 32×32 and 64×64 large blocks.

To simultaneously describe both location information and partition depth information within the encoding domain features of the CDU structure, this paper employs a value mapping approach to transform them into graph features. The mapping method of CU is illustrated in Fig. 4b. The number in the block denotes the pixel value of the map.

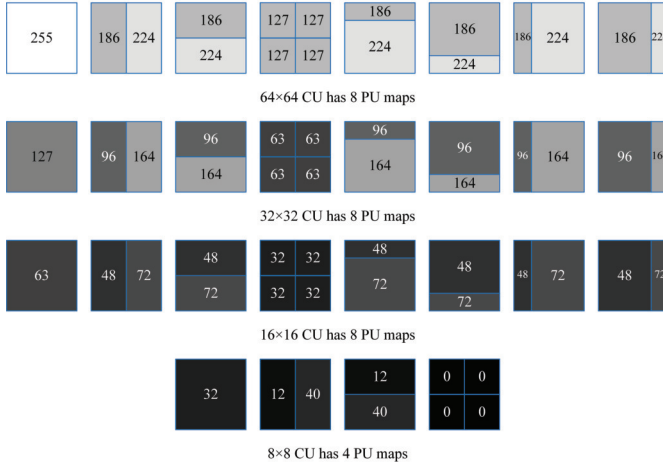
2) *TU Map*: TU (Transform Unit) is the fundamental unit for transformation and quantization in the HEVC encoding process. TU partitioning starts from the root node of a CU block using a quadtree structure, with only symmetric partitioning allowed. TU sizes include 64×64 , 32×32 , 16×16 , 8×8 , and 4×4 . Due to their smaller partitioning granularity than CUs, TUs can more accurately capture fine-grained texture details in input video frames.

Similarly, 100 samples are selected and conducted frequency histogram analyses of TU partitioning for both adversarial and clean samples, as depicted in Fig. 5a. It is observable that adversarial samples exhibit higher frequencies in smaller TU sizes, such as 4×4 , 8×8 , and 16×16 . Conversely, there is a marked decrease in frequency for larger TU sizes of 32×32 and 64×64 , indicating that the encoder perceives fewer smooth regions in adversarial samples than their clean counterparts. The mapping methods for TU are shown in Fig. 5b.

3) *PU Map*: PU (Prediction Unit) represents the fundamental inter- and intra-frame prediction unit in HEVC encoding. Like TU, PU partitioning also begins with the CU root node. For CU blocks with sizes of 16×16 or larger, eight distinct PU partitioning modes exist: four symmetric and four asymmetric.



(a) Histogram of PU Frequency Distribution



(b) PU maps

Fig. 6. Histograms depicting the frequency distributions of PU blocks generated by different samples in HEVC encoding.

Symmetric modes allow a $2N \times 2N$ CU block to be split into a $2N \times 2N$ sub-block, four $N \times N$ sub-blocks, two horizontally-oriented $N \times 2N$ sub-blocks, or two vertically-oriented $2N \times N$ sub-blocks. The four asymmetric modes enable a $2N \times 2N$ CU block to be divided horizontally into a $1/2N \times 2N$ and a $3/2N \times 2N$ sub-block, horizontally into a $3/2N \times 2N$ and a $1/2N \times 2N$ sub-block, vertically into a $2N \times 3/2N$ and a $2N \times 1/2N$ sub-block, or vertically into a $2N \times 1/2N$ and a $2N \times 3/2N$ sub-block. For CU blocks of size 8×8 , only symmetric partitioning modes are applicable.

The frequency distribution histograms of PU partitioning for both adversarial and clean video samples are shown in Fig. 6a. Adversarial samples are more inclined towards smaller PU blocks. In comparison, clean samples utilize larger PU blocks. This phenomenon can be attributed to two factors. Firstly, introducing adversarial perturbations renders the image texture more complex, necessitating smaller block sizes for improved prediction accuracy. Secondly, the higher prevalence of smaller blocks in the CU partitioning confines the PU division to even smaller units. Mapping PU values presents a more intricate challenge. In this regard, our work draws inspiration from the mapping approach developed by Professor Zhao's research [23] team for addressing PU mapping. The mapping methods for PU are shown in Fig. 6b.

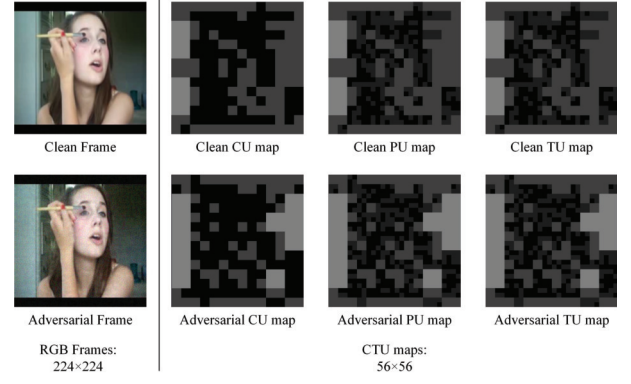


Fig. 7. Illustrations of clean and adversarial frames, together with their corresponding CDU maps.

Fig. 7 shows the CDU map images obtained by encoding clean and adversarial samples using HEVC under the same QP constraints. It can be seen that the division of CDUs in adversarial samples is finer, with more small-sized CDU blocks used.

IV. PROPOSED ALGORITHM

From the preceding analysis, it can be inferred that semantic information within video samples is strongly associated with CDU partitioning in the compression domain. However, the introduction of adversarial noise disrupts this association. Despite the decreased correlation with semantic features, the CDU partitioning in adversarial samples remains highly correlated with their texture features. Consequently, more than simple texture matching-based detection is needed to identify adversarial traces reliably.

In response, this paper proposed a dual-stream video adversarial frame detection network. The first step of this process involves purifying the input sample from adversarial perturbation, thereby mitigating the texture artifacts induced by adversarial noise. This purification process is designed to ideally restore the original frame's semantic features. Then, the consistency between the semantic information of the purified sample and the CDU maps is calculated. The consistency of adversarial samples is poor, while the consistency of clean samples is high. The training and inference process of our method is illustrated in Fig. 8.

A. Training of Preprocess:

During the preprocessing stage, the role of the adversarial perturbation purifier is to remove adversarial noise from the input sample as much as possible to restore its original RGB semantic space information. To achieve this, an Adversarial Purifier (AP) is employed based on a denoising autoencoder architecture. The AP consists of a downsampling structure E and an upsampling structure G . During training, for a clean video frame F , corresponding adversarial frame F' is generated by applying a specific adversarial perturbation. For sample pair $\langle F, F' \rangle$, E performs feature representations l and l' , while G generates images $\hat{F}$ and $\hat{F}'$. It is desired that AP be more sensitive to adversarial noise; hence, the loss function L_{re}

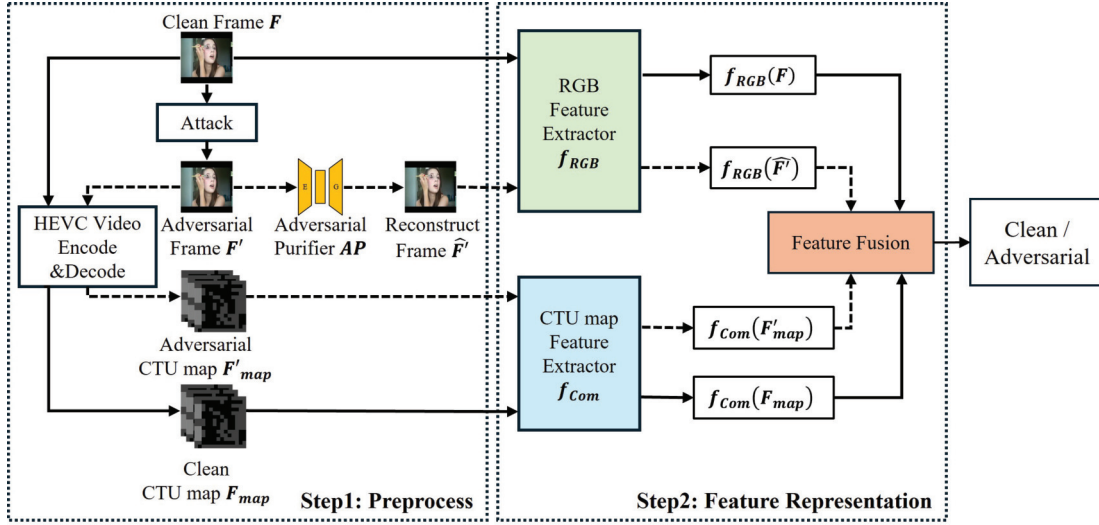


Fig. 8. Training process of our proposed method.

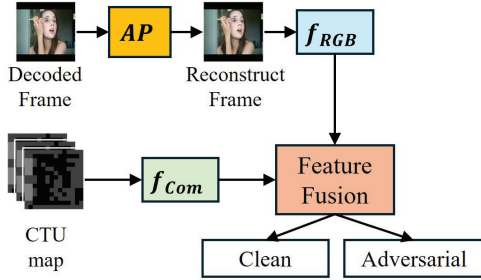


Fig. 9. Inference process of our proposed method.

is given by Eq. (13). Where $MSE(\cdot)$ denotes Mean Squared Error Loss, λ_1 and λ_2 are hyper-parameters.

$$L_{re} = MSE(l, l') + \lambda_1 \cdot MSE(F, \hat{F}) + \lambda_2 \cdot MSE(F, \hat{F}') \quad (13)$$

Subsequently, for frame pair F and F' , HEVC video encoding and decoding operations are performed separately on each. The CU map, PU map, and TU map are extracted from the respective frames, and these three grayscale maps are concatenated into a single three-channel image, CDU map, with dimensions equal to 1/16 of the original image size. The map corresponding to F is denoted as F_{map} , and the map corresponding to F' is denoted as F'_{map} .

$$CDU_{map} = [CU_{map}, PU_{map}, TU_{map}] \quad (14)$$

Thus, in the preprocessing stage, a quadruplet $\langle F, \hat{F}', F_{map}, F'_{map} \rangle$ is adopted.

B. Training of Feature Representation:

For the quadruple input generated during the preprocessing stage, where F and F' are RGB images, RGB feature extractor f_{RGB} is adopted to perform feature extraction and normalization on each, yielding $f_{RGB}(F)$ and $f_{RGB}(\hat{F}')$, the shape of RGB features is 1×512 . For the compression domain images F_{map} and F'_{map} , the CDU map feature extractor f_{Com} is used to extract and normalize their features, resulting in

$f_{Com}(F_{map})$ and $f_{Com}(F'_{map})$, the shape of compression features is 1×512 . Cosine similarity ($CosSim(\cdot)$) is utilized to measure the similarity between two distributions. In the feature representation phase, our objectives are as follows: firstly, the representations of clean samples' RGB and compression domain features should be consistent. Thus match loss ML_{clean} between F and F_{map} is proposed in Eq. (15).

$$ML_{clean} = CosSim(f_{RGB}(F), f_{Com}(F_{map})) \quad (15)$$

Secondly, the RGB features of reconstructed adversarial samples should align with the compression domain feature representation of clean samples. The match loss ML_{re} between reconstruct adversarial sample $\hat{F}'$ and clean CDU map F_{map} is shown in Eq. (16).

$$ML_{re} = CosSim(f_{RGB}(\hat{F}'), f_{Com}(F_{map})) \quad (16)$$

Lastly, the RGB features of reconstructed adversarial samples should exhibit a discrepancy compared to the compression domain feature representation of adversarial samples. The match loss ML_{adv} between reconstruct adversarial sample $\hat{x}'$ and adversarial CDU map x'_{map} is shown in Eq. (17).

$$ML_{adv} = CosSim(f_{RGB}(\hat{F}'), f_{Com}(F'_{map})) \quad (17)$$

Ultimately, during feature representation training, the matching degree loss L_{match} calculation is represented by the combination of the three losses above, with its computation formula given by Eq. (18). Where λ_{re} and λ_{adv} are hyper-parameters.

$$\begin{aligned} L_{match}(F, \hat{F}', F_{map}, F'_{map}) \\ = ML_{clean} + \lambda_{re} \cdot ML_{re} - \lambda_{adv} \cdot ML_{adv} \end{aligned} \quad (18)$$

Through optimization of the above loss function, the feature representation phase is expected to yield effective extraction of RGB and compression domain features, enabling efficient calculation and mapping of their match degrees to support subsequent feature fusion and adversarial detection classification.

Subsequently, feature fusion is carried out from RGB features and CDU map features, with the fusion method

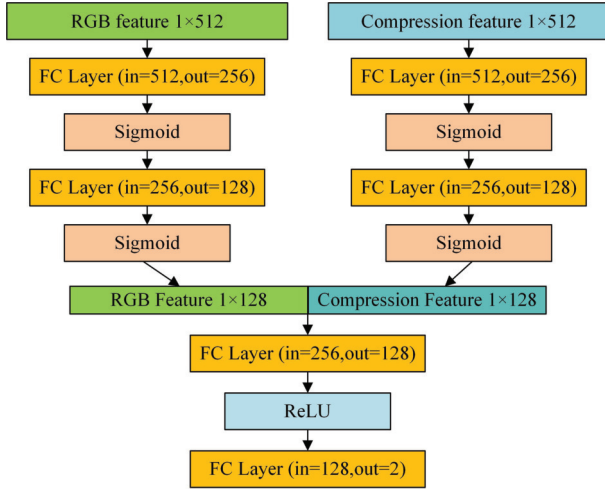


Fig. 10. Feature fusion and classification architecture.

illustrated as depicted in Fig. 10. In the final stage of optimizing classification errors, since the magnitude of frames with adversarial perturbations added in sparse video adversarial samples is significantly lower than that of frames without such perturbations, directly computing cross-entropy loss would lead to class imbalance issues. To address this, focus loss L_{cls} is utilized for optimization, with its optimization function presented in Eq. (19).

$$L_{cls}(p_t) = -\alpha(1 - p_t)^\gamma \log(p_t) \quad (19)$$

Here, p_t represents the confidence that a sample belongs to its actual label, with adversarial frames labeled as 1 and clean frames labeled as 0. α is a balancing parameter designed to mitigate the issue of unequal numbers between adversarial and clean frames. In this paper, it is set to 0.25. γ serves as a hyper-parameter adjusting the balance between easily trainable and difficult-to-train samples during training, with a value of 2 in this study. Ultimately, in the feature representation stage, the loss function L employed during training is a weighted sum of the matching loss L_{match} and the classification loss L_{cls} , where λ_{match} denotes the hyper-parameter for optimization.

$$L = L_{cls} + \lambda_{match} \cdot L_{match} \quad (20)$$

C. Voting Strategy for Video Level Detection:

Following the acquisition of frame-level adversarial detection results, video-level detection can be achieved by implementing a voting strategy. Given a video sequence $V = \{F_1, F_2, \dots, F_N\}$, adversarial frame detection is performed on each frame F_i within it, yielding detection result r_i . The discrimination strategy for the V is expressed by Eq. (21). The V is deemed an adversarial video when the number of detected adversarial frames within it exceeds threshold th . The value of th is determined by the training process, and the th that performs best in the training set will be selected as the threshold for testing.

$$V = \begin{cases} \text{Adversarial} & \left(\sum_{i=1}^N r_i \right) > th \\ \text{Clean} & \left(\sum_{i=1}^N r_i \right) \leq th \end{cases} \quad (21)$$

TABLE I
MODEL ACCURACY

Model	UCF-101	HMDB-51	Kinetics-400
I3D	94.26%	70.13%	74.80%
R(2+1)D	95.56%	74.90%	-
SlowFast	95.77%	75.62%	-

V. EXPERIMENTS AND ANALYSIS

In this section, our proposed method in this paper is put to the test, undergoing comparison against state-of-the-art adversarial detection approaches across various adversarial attack methods and target models.

A. Experiment Setup

1) *Target Models*: Three models, I3D [20], R(2 + 1)D [24], and SlowFast [25], were used as target video action recognition models, and their accuracies on public video datasets UCF-101 [26], HMDB-51 [27] and Kinetics-400 are shown in Table I. For Kinetics-400, only videos with resolution greater than 720p are selected for test.

2) *Datasets*: When conducting adversarial detection, for the target model and target public video dataset, four types of white-box attacks were selected: FGSM [21], PGD [22], SparseAttack (SA) [9], and Anti Codec Attack (ACA) [4], as well as five types of black-box attacks: Cross Model Attack (CMA) [6], Temporal Translation Attack (TTA) [5], SVA [28], AstFocus [7], and CLVA [8]. Among them, SA, SVA, AstFocus, and CLVA are sparse attacks, while the rest are global attacks. The parameter settings for various attacks are shown in Table II. The number of generated video adversarial samples are shown in Table III, 70% of which is used for training, and 30% for testing.

3) *Video Encoding*: In this experiment, the video coding method adopted is HEVC, with the encoder being libx265, the decoder being HM, a GOP length of 4 for encoding, and a fixed QP of 20.

4) *Metrics*: In this experiment, the adversarial detection algorithm is measured using the Area Under the Receiver Operating Characteristic Curve (AUROC) score.

5) *Detection Methods*: For our proposed video adversarial detection method, adversarial perturbation purifiers and feature representations adopt distributed training. For AP, λ_1 is set to 0.5 during training, λ_2 is set to 1, and the learning rate is 0.001. In the feature representation stage, both the f_{RGB} and the f_{Com} use the feature extractor of the pre-trained ResNet18 [29] network on ImageNet2012 [30]. The learning rate of f_{RGB} is 0.001, the learning rate of f_{Com} is 0.0005, λ_{re} is 2, and λ_{adv} is 1. λ_{match} is set to 0.25 when training the classifier.

In the experiment, four SOTA adversarial detection methods, AdvIT [31], SmsNet [32], GraN [33], and ADDNP [10], were selected for comparison.

AdvIT is a video adversarial detection method based on optical flow thresholding. In this experiment, we use FlowNet 2.0 [34] implemented in mmflow to simulate the generation of optical flow. A separate threshold is required for each

TABLE II
ATTACK PARAMETERS USED IN THIS PAPER

Attack Type	Attack Method	Sparse Attack	Attack Parameters
White-box	FGSM [21]	No	Epsilon=8/255
	PGD [22]	No	Epsilon=8/255, step size=2/255, iteration number=10
	SA [9]	Yes	Epsilon=8/255, step size=2/255, iteration number=50, number of attacked frames=4
	ACA [4]	No	Epsilon=8/255, step size=2/255, iteration number=50, video encoder=HEVC
Black-box	CMA [6]	No	Image model=ResNet-101, epsilon=16/255, feature block=block2, step size=0.0005, iteration number=60
	TTA [5]	No	Epsilon=16/255, step size=1.6/255, iteration number=10
	SVA [28]	Yes	LSTM hidden dims=128, frame model=ResNet18, epsilon=0.05, max query number=30000, NES population number=48, NES search variance=0.001
	AstFocus [7]	Yes	Untarget attack, epsilon=0.05, max query number=30000
	CLVA [8]	Yes	Untarget attack, epsilon=0.05, max query number=30000, number of attacked frames=6

TABLE III
NUMBER OF ADVERSARIAL SAMPLES DURING EXPERIMENTS

Attacks	UCF-101			HMDB-51			Kinetics-400
	I3D	R(2+1)D	SlowFast	I3D	R(2+1)D	SlowFast	I3D
FGSM	1405	1076	1010	1150	1258	1365	1200
PGD	1142	1142	1141	1292	1469	1149	1159
SA	1178	1444	1470	1259	1353	1080	1346
ACA	1152	1471	1173	1472	1231	1059	1211
CMA	1447	1158	1032	1102	1070	1241	1159
TTA	1079	1279	1067	1038	1267	1400	1258
SVA	1421	1193	1025	1143	1491	1390	1223
AstFocus	1236	1578	1119	1115	1228	1397	1159
CLVA	1048	1116	1244	1512	1412	1539	1687
Total	11108	11457	10281	11083	11779	11620	11402

adversarial attack method. First, the optimal threshold is found in the training set and then applied to the test set for result evaluation.

SmsNet is a statistical-based adversarial detection method. In this experiment, we select the five-layer network structure proposed in its original paper as the benchmark model. The input image size is 224×224 , the initial learning rate is 0.001, the batch size is 32, and the model is trained for 100 epochs using the SGD optimizer.

GraN is an adversarial detection method based on the distance of intermediate layer features. During the training of this network, following the original paper, we applied Gaussian smoothing with a kernel size of 5 to process the input images. We used I3D as the target gradient extraction network, extracting 50 layers of gradient information. Then, the L1 distance between each layer of gradients was calculated. Finally, a fully connected layer with an input size of 50 and an output size of 1 was used as the logistic regression layer for classification. The optimal threshold was first selected in the training set and then used for evaluation in the test set.

ADDNP is an adversarial detection method based on image reconstruction and changes in classification by the target network. Therefore, adversarial samples generated for different target networks require detection using their corresponding target networks. Following the original paper, we use the NRP network for image reconstruction. Then, these reconstructed

images are reassembled into video sequences and fed into the corresponding target networks for inference. The NRP network [11] for reconstruction is pre-trained on ImageNet2012, and its network structure follows the one provided in the open-source code. The threshold judgment is then made by calculating the output cosine similarity between the reconstructed and original images. The optimal threshold is first identified in the training set and then applied for evaluation in the test set.

B. Detection in White-Box Attacks

The comparison experiments of adversarial detection against white-box attacks on the UCF-101 and HMDB-51 datasets are shown in Table IV. It can be seen that AdvIT performs well in most cases but poorly in the Slowfast FGSM attack, with only 0.6859. It is presumably because Slowfast uses a dual-channel recognition model. In the single-step iterative FGSM attack, the generated optical flow changes are small and difficult to distinguish. SmsNet performs poorly in SA attacks with only 0.5 to 0.6 but performs well in other methods. This phenomenon is because the sparse attack of SA leads to a minor deviation in the distribution of statistical quantities, making it difficult to detect. Similar situations also occur in ADDNP, where their detection performance in SA is significantly lower than that of other adversarial samples. Sparse attacks are a complex problem for previous adversarial detection methods. The detection performance of GraN in SA is significantly affected by the target model. Its SA detection on I3D is only 0.6434, but it achieves 0.9418 on R(2 + 1)D and even reaches 0.9608 in GraN. Since GraN detects based on changes in intermediate feature layers, resulting in the intermediate features of I3D are more challenging to distinguish. In contrast, the feature dimensions of 2D convolutions in R(2 + 1)D and SlowFast are lower and more easily distinguishable.

Our method has achieved the best results in most experiments. Even in the experiments that did not achieve the best, such as the ACA detection of R(2 + 1)D on UCF-101, it still reached a performance of 0.9998, which is only 0.0002 lower than the best method. In the other experiments where our method did not achieve the best, it was still the second best, and the gap between our method and the optimal detection

TABLE IV
AUROC SCORES OF WHITE-BOX ADVERSARIAL DETECTION. THE BEST RESULTS ARE BOLD

Models	Methods	UCF-101					HMDB-51				
		FGSM	PGD	SA	ACA	Average	FGSM	PGD	SA	ACA	Average
I3D	AdvIT [31]	0.9845	0.9598	0.8869	0.9799	0.9528	0.7327	0.9855	0.6867	0.8508	0.8139
	SmsNet [32]	0.9839	0.9987	0.5527	0.9831	0.8796	0.9824	0.9320	0.5009	0.9756	0.8477
	GraN [33]	0.9437	0.9048	0.6434	0.9509	0.8607	0.9119	0.9809	0.5765	0.9568	0.8565
	ADDNP [10]	0.9922	0.9894	0.6932	0.9861	0.8896	0.9951	0.9959	0.6382	0.9813	0.9026
	Ours	1.0000	1.0000	0.9999	1.0000	1.0000	0.9945	0.9886	0.9868	0.9956	0.9914
R(2+1)D	AdvIT [31]	0.8913	0.9988	1.0000	1.0000	0.9725	0.5185	0.9947	1.0000	0.9673	0.8701
	SmsNet [32]	1.0000	1.0000	0.6289	0.9992	0.9070	1.0000	0.8313	0.5156	0.9855	0.8331
	GraN [33]	0.9881	0.9829	0.9418	0.9927	0.9764	0.9893	0.9822	0.9515	0.9887	0.9779
	ADDNP [10]	0.9884	0.9933	0.6907	0.9973	0.9174	0.9837	0.9805	0.6758	0.9564	0.8991
	Ours	1.0000	1.0000	1.0000	0.9998	1.0000	1.0000	1.0000	0.9999	1.0000	1.0000
SlowFast	AdvIT [31]	0.6859	1.0000	1.0000	0.8701	0.8890	0.5199	0.9998	1.0000	0.9785	0.8746
	SmsNet [32]	0.9709	0.7855	0.5448	0.9447	0.8115	1.0000	0.9407	0.5037	0.8784	0.8307
	GraN [33]	0.8794	0.9354	0.9608	0.9617	0.9343	0.9196	0.9859	0.9567	0.9705	0.9582
	ADDNP [10]	0.9903	0.9934	0.6970	0.9902	0.9177	0.9903	0.9934	0.6976	0.9902	0.9179
	Ours	1.0000	1.0000	1.0000	0.9998	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

method was within 0.01. This result proves that our method has good universality in detecting white-box attacks and does not change with different attack patterns.

Our method has demonstrated excellent detection capabilities against different white-box attacks, achieving the best or second-best detection results in all experiments. This result proves that our method is a more robust video adversarial detection method that is insensitive to attack patterns and attack sparsity levels.

C. Detection in Black-Box Attacks

The experimental results of adversarial detection against black-box attacks are shown in Table V, where CMA and TTA are transfer-based global attacks, while SVA, AstFocus, and CLVA are query-based sparse attacks. Our proposed method achieves the best detection performance in most experiments and only lags behind the best SmsNet by 0.005 in the CMA detection on I3D HMDB-51.

The AdvIT method performs well in detecting TTA but poorly in other detections. The AdvIT assumption is that the consistency calculated from adversarial samples will be lower than that of clean samples. TTA is an iterative attack that significantly affects the optical flow continuity during the attack process, so AdvIT can effectively detect it. However, CMA is a video adversarial sample generated by attacking the image model. Considering that the difference between consecutive frames is minimal for the image model, and the perturbations between frames are also very similar, the resulting CMA adversarial samples do not significantly differ from clean samples in consistency, resulting in the ineffectiveness of AdvIT. Additionally, AdvIT performs poorly in other black-box sparse attacks. This is because the selection of keyframes is involved in black-box sparse attacks, which leads to different keyframes being selected for different samples, resulting in

different patterns of optical flow changes. Therefore, it is difficult for AdvIT to detect them.

SmsNet and GraN are detection methods that are highly sensitive to the target model and dataset. For example, when detecting CMA, SmsNet only achieves a performance of 0.6945 on I3D-UCF101 but achieves a perfect score of 1.0000 in all other cases. When detecting TTA, the performance of SmsNet also fluctuates significantly with changes in the target model and dataset, achieving only 0.6534 on I3D-HMDB51 but 0.9961 on R(2 + 1)D-UCF101. This result demonstrates that SmsNet lacks generalization across different scenarios. SmsNet is a detection method based on statistical measures, and the statistical values in different models and datasets can vary significantly, making it difficult to guarantee the stability of detection performance. SmsNet and GraN also exhibit significant performance fluctuations when detecting black-box sparse attacks. For example, when GraN detects CLVA, it can achieve AUROC of 0.8967 and 0.8992 on R(2 + 1)D but performs poorly on other models. Relying too much on statistics for discrimination in adversarial detection can lead to significant fluctuations in detection performance when faced with different models and datasets due to differences in the underlying data distribution. This result means that statistic-based is an incomplete approach to adversarial detection.

ADDNP performs well in CMA and TTA and exhibits stable performance across different models and datasets. For example, in UCF-101, the detection performance for CMA is above 0.99, and the detection performance for TTA is above 0.97. This result demonstrates that ADDNP performs well against global attacks. However, its ability to detect sparse attacks could be better. For example, in the detection of SVA, it only achieves 0.6224 on I3D-UCF101, and the detection of AstFocus and CLVA is only 0.5975 and 0.6239, respectively. This result indicates that ADDNP cannot detect sparse attacks. Since ADDNP purifies input samples through

TABLE V
AUROC SCORES BLACK-BOX ADVERSARIAL DETECTION PERFORMANCE. THE BEST RESULTS ARE BOLD

Models	Methods	UCF-101						HMDB-51					
		CMA	TTA	SVA	AstFocus	CLVA	Average	CMA	TTA	SVA	AstFocus	CLVA	Average
I3D	AdvIT [31]	0.5000	0.9497	0.6490	0.4831	0.5000	0.6164	0.1401	0.9626	0.4100	0.3401	0.4087	0.4523
	SmsNet [32]	0.6945	0.8490	0.6945	0.5845	0.6846	0.7014	1.0000	0.6534	0.6857	0.5935	0.6856	0.7236
	GraN [33]	0.7273	0.6559	0.7273	0.5877	0.7201	0.6837	0.9642	0.7333	0.7368	0.7359	0.7366	0.7814
	ADDNP [10]	0.9993	0.9760	0.6224	0.5975	0.6239	0.7638	0.9932	0.9725	0.5886	0.6029	0.5533	0.7421
	Ours	1.0000	0.9997	0.8139	0.9967	0.9858	0.9592	0.9950	0.9923	0.8580	0.9542	0.9778	0.9555
R(2+1)D	AdvIT [31]	0.0736	0.9967	0.6885	0.5483	0.6859	0.5986	0.3748	0.9790	0.7164	0.4775	0.7143	0.6524
	SmsNet [32]	1.0000	0.9961	0.8052	0.6164	0.7955	0.8426	1.0000	0.7279	0.7784	0.5489	0.7690	0.7648
	GraN [33]	0.9898	0.9869	0.9056	0.5501	0.8967	0.8658	0.9968	0.9907	0.9058	0.6221	0.8992	0.8829
	ADDNP [10]	0.9922	0.9802	0.6220	0.5637	0.5903	0.7497	0.9873	0.9268	0.5635	0.6185	0.5497	0.7292
	Ours	1.0000	1.0000	0.9154	0.9273	0.9455	0.9576	1.0000	1.0000	0.9146	0.9588	0.9777	0.9702
SlowFast	AdvIT [31]	0.1910	0.9843	0.6954	0.5828	0.6918	0.6291	0.2698	0.9963	0.6944	0.5665	0.6887	0.6431
	SmsNet [32]	0.9999	0.5593	0.6632	0.5356	0.6601	0.6836	1.0000	0.6563	0.6587	0.7021	0.6538	0.7342
	GraN [33]	0.9383	0.6350	0.8163	0.6208	0.8151	0.7651	0.9566	0.9461	0.7659	0.6899	0.7634	0.8244
	ADDNP [10]	0.9986	0.9840	0.5984	0.6556	0.8186	0.8110	0.9945	0.9525	0.5520	0.6855	0.5787	0.7527
	Ours	1.0000	0.9999	0.9006	0.9364	0.9663	0.9606	1.0000	1.0000	0.9154	0.9453	0.9688	0.9659

the NRP network and then uses the target model to measure the output consistency before and after purification, when the attack is sparse, NRP has difficulty accurately purifying the perturbations on each frame, making it difficult to set the threshold for the output consistency of the template model, ultimately leading to poor detection results.

In the context of video adversarial detection, introducing sparsity in the temporal dimension poses a significant challenge for previous methods. When video adversarial attacks adopt similar attack patterns to image attacks, such as ACA, CMA, and TTA, previous adversarial detection methods can effectively detect them. However, introducing sparsity in the temporal dimension, a key characteristic of video adversarial samples, poses a significant challenge. Most previous methods fail to effectively detect sparse video adversarial samples, which underscores the limitations of these methods in the face of evolving attack patterns.

Our method, starting from the consistency of compression domain features and RGB features, comprehensively considers the effectiveness of spatial texture features (CU and TU maps) and temporal motion features (PU maps) in video adversarial samples. The proposed video adversarial detection method exhibits good performance and generalization across attack patterns, target models, and datasets.

D. High-Resolution Video Test

Considering that UCF-101 and HMDB-51 are low-resolution datasets, we also tested the detection performance of adversarial videos at high resolutions using the Kinetics-400 dataset. Specifically, we selected 1000 videos with a resolution greater than 720p from the test set of the Kinetics-400 dataset, and the I3D network correctly identified all 1000 videos. We then subjected them to adversarial attacks to generate adversarial samples. During the attack process, all adversarial perturbations were directly added to the videos at their original

size through a differentiable resize operation, ensuring that the generated adversarial videos would not be affected by the input size requirements of the target model.

The results of the experiment are shown in Table VII. Our method still maintains high performance in high-resolution scenarios, consistently detecting different types of attacks, as seen with low-resolution datasets. High-resolution data contains richer encoded information compared to low-resolution data. For instance, features such as CDU partition patterns will be more pronounced, and RGB pixel domain features will be more apparent, thus ensuring that our method performs well when dealing with high-resolution videos.

However, high-resolution adversarial videos have a significant impact on the GraN method. In detecting CMA, GraN performs well on low-resolution data but poorly on high-resolution data. This result is because GraN is a detection method based on the statistics of intermediate gradients in the model. When the video resolution increases, the dimensionality of the model's intermediate feature gradients will rise, leading to a more complex and random distribution, which in turn causes GraN to fail in detecting effective features.

E. Generalization Performance

In the real-world adversarial detection scenarios, defenders cannot know the adversarial attack methods used by attackers, so the generalization ability of adversarial detection algorithms is particularly important. In this experiment, FGSM is selected as the training sample for global attack, and SA is selected as the training sample for sparse attack. The adversarial detection is trained on FGSM and SA, and the performance of different adversarial detectors under other attacks is measured.

The experimental results are shown in Table VI. Since AdvIT and ADDNP use thresholds for judgment, the F1 score is used as a measure in the generalization experiments. When trained on FGSM, AdvIT achieved the highest accuracy

TABLE VI
F1 SCORES OF GENERALIZATION EXPERIMENTS. THE TRAINING SAMPLES ARE GENERATED ON THE I3D
MODEL AND UCF101 DATASET, AND THE BEST RESULTS ARE BOLD

Train Attack	Methods	Test Attack									
		FGSM	PGD	SA	ACA	CMA	TTA	SVA	AstFocus	CLVA	Average
FGSM	AdvIT [31]	-	0.5000	0.4849	0.4975	0.4975	0.5025	0.5000	0.5000	0.4937	0.4970
	SmsNet [32]	-	0.9684	0.5777	0.5606	0.5034	0.7438	0.5049	0.4887	0.4359	0.6479
	GraN [33]	-	0.9050	0.4572	0.9539	0.7912	0.6623	0.8291	0.7693	0.7941	0.7703
	ADDNP [10]	-	0.9071	0.0952	0.8439	0.9744	0.8439	0.0000	0.0198	0.0198	0.4630
	Ours	-	0.9943	0.9057	0.9973	1.0000	0.9963	0.8806	0.8998	0.7767	0.9313
SA	AdvIT [31]	0.4975	0.8719	-	0.8869	0.4869	0.4975	0.5000	0.5000	0.4582	0.6573
	SmsNet [32]	0.6130	0.6233	-	0.5234	0.5010	0.5748	0.5021	0.4996	0.4748	0.5284
	GraN [33]	0.6130	0.6233	-	0.5234	0.5010	0.5748	0.5021	0.5021	0.4748	0.5288
	ADDNP [10]	0.9208	0.9314	-	0.9261	0.9519	0.9208	0.2059	0.2131	0.2276	0.6253
	Ours	0.9912	0.9800	-	0.9782	0.9890	0.9825	0.9020	0.9822	0.7611	0.9393

TABLE VII
THE PERFORMANCE ON HIGH-RESOLUTION DATASETS WAS TESTED,
WITH VIDEOS FROM THE KINETICS-400 DATASET WITH A
RESOLUTION GREATER THAN 720P SELECTED
AS TEST EXAMPLES

Attack Methods	Detection Methods				
	AdvIT	SmsNet	GraN	ADDNP	Ours
FGSM	0.7583	0.7112	0.5688	0.9969	0.9968
PGD	0.8010	0.7274	0.8400	0.9938	1.0000
SA	0.7792	0.5286	0.7169	0.6827	0.9999
ACA	0.6022	0.8717	0.9610	0.9850	1.0000
CMA	0.2778	0.5196	0.4128	0.9488	1.0000
TTA	0.8128	0.5764	0.6789	0.9980	0.9997
SVA	0.5109	0.4622	0.7092	0.4766	0.8532
AstFocus	0.3379	0.4548	0.5014	0.4554	0.9999
CLVA	0.4306	0.4605	0.6304	0.5109	0.9786
Average	0.5901	0.5903	0.6688	0.7831	0.9809

of 0.5025 in detecting TTA and the lowest of 0.4849 in detecting SA. It is presumably because AdvIT is a detection method based on optical flow information. When the attack mode changes, the pattern of perturbed optical flow will also change, resulting in poor performance in other methods. For SmsNet, it performed best in PGD, achieving 0.9684, with the second highest score among the five methods. However, its performance was poor in other methods, such as only 0.4359 in CLVA. GraN performed well in PGD, ACA, and CLVA, and it was the best among the five methods in CLVA. As for ADDNP, its detection performance highly depends on the video attack method. Since FGSM is a global attack method, it performed well in tests such as PGD, ACA, CMA, and TTA, achieving a second-best score of 0.9744 in CMA. However, its performance was very poor in sparse attacks, making it unusable. This means that ADDNP cannot maintain good generalization when the attack mode changes.

Our method demonstrates good performance when tested against different unknown attacks. It achieves optimal or

second-best performance in all detections, with an average detection score of 0.9313, indicating that our method can effectively detect unknown attacks when trained on FGSM.

When trained on SA, the situation has undergone some changes. AdvIT performs well on PGD and ACA, but its performance is generally average in other detection methods. PGD, ACA, and CMA are all based on iterative attacks, while TTA also uses iterative attacks but smooths out the differences in optical flow by using temporal dimension translation during the attack process. In contrast, FGSM with single-step iteration, SVA, AstFocus and CLVA with black-box querying have different attack patterns, resulting in significant performance differences. Moreover, the SmsNet and GraN trained on SA have difficulty achieving good detection results in other attack methods. SmsNet and GraN are based on statistical metrics, and when the attack pattern changes, the detection performance will decrease significantly. The ADDNP method performs well in global attacks when trained on SA, which is understandable because each frame is reconstructed during detection, and the output of the reconstructed adversarial video samples will vary greatly. Therefore, training on SA can also successfully detect global attacks. However, for methods such as SVA, AstFocus, and CLVA, where the position of the attacked frame is uncertain, ADDNP has difficulty achieving good detection results.

Our method can also achieve good performance when trained using SA, achieving the best detection results among all attack methods. In particular, it outperforms the detection results obtained from training on FGSM on SVA and AstFocus, with leading margins of 0.0214 and 0.0824, respectively. While there is a slight decrease in other methods, the overall average detection performance still reaches 0.9393, slightly better than training on FGSM.

In summary, our proposed method has the best generalization ability for video adversarial detection compared to other methods, and can meet the requirement of training with low-cost attack samples (such as FGSM and SA) to

TABLE VIII
THE IMPACT OF VIDEO ENCODING GOP SIZE ON DETECTION

GOP size	Attack Methods				
	SA	ACA	AstFocus	CLVA	Average
4	0.9999	1.0000	0.9967	0.9858	0.9956
8	0.9858	0.9954	0.9742	0.9858	0.9853
16	0.9698	0.9946	0.9706	0.9753	0.9776
32	0.9553	0.9887	0.9701	0.9543	0.9671

TABLE IX
THE IMPACT OF VIDEO ENCODING QP ON DETECTION

QP	Attack Methods				
	SA	ACA	AstFocus	CLVA	Average
20	0.9999	1.0000	0.9967	0.9858	0.9956
25	0.9999	0.9999	0.9967	0.9855	0.9955
30	0.9885	0.9963	0.9932	0.9805	0.9896
35	0.9546	0.9448	0.9698	0.9785	0.9619
40	0.9225	0.9086	0.9120	0.9664	0.9274
45	0.8972	0.9002	0.8547	0.9283	0.8951
50	0.8542	0.8993	0.8054	0.9072	0.8665

defend against high-cost adversarial attacks (such as CLVA and AstFocus).

F. Robustness Evaluation

In real-world scenarios, the parameter settings of the H.265 encoder and specific video processing techniques can affect the performance of adversarial detection. This section will test the robustness of the proposed method's detection performance from group of pictures (GOP) size changing, quantization parameter (QP) changing, and general video processing.

1) *Video Encoding Parameters*: In robustness of video encoding parameters testing, we selected the I3D model as the target model and conducted tests on the UCF-101 dataset. Two types of white-box attacks, SA and ACA, as well as two types of black-box attacks, AstFocus and CLVA, were used as detection targets.

Table VIII demonstrates the adversarial detection performance of this method under different GOP sizes. It can be observed that as the GOP size increases, there is a slight decrease in the detection performance of this method. For instance, in SA detection, when the GOP size is 32, the detection AUROC scores decreased by 0.0446. On average, the detection performance decreased by 0.0285 for these four attack methods.

Table IX illustrates the changes in the detection performance of this method as QP increases, which corresponds to a decrease in the quality of encoded videos. When QP is below 35, the performance of the detection method declines relatively slightly, with a decrease of 0.0552 in ACA detection and only a 0.0073 decrease in CLVA. However, when QP increases, the detection performance declines more noticeably. For example, when QP rises to 50, the detection for AstFocus drops to 0.8054.

TABLE X
THE IMPACT OF VIDEO PROCESSING ON DETECTION PERFORMANCE

Video Processing	Processing Parameter	UCF-101	HMDB-51	Kinetics-400 (720p+)
Frame deletion	First frame	1.0000	0.9999	0.9999
	Last frame	0.9998	0.9999	0.9999
	First+Last frames	0.9967	0.9983	0.9912
Frame rate conversion	25fps→50fps	1.0000	0.9999	0.9999
	25fps→12fps	1.0000	0.9999	0.9999
Video compression	QP20→25	0.9999	0.9999	0.9999
	QP20→30	0.9785	0.9854	0.9954
	QP20→35	0.9542	0.9499	0.9766
Video transcoding	MPEG4→HEVC	0.9742	0.9887	0.9903
	H.264→HEVC	0.9779	0.9745	0.9896
Social media	Weibo	0.9265	0.9115	0.9327
	WeChat	0.8977	0.9264	0.9339

Then, a high QP corresponds to the failure of adversarial attacks. In the experiments, when QP is increased above 35, many perturbations in the adversarial samples have been eliminated. Our experiments also observed that under high QP, the adversarial samples that were misclassified as clean samples had essentially lost their attacking capability.

2) *Video Processing*: In the video processing robustness experiment, we considered various methods that might occur in real-world scenarios, including frame deletion, frame rate conversion, video compression, video transcoding, and social media transmission. Since these video processing methods involve multiple video encodings, which can severely impair the effectiveness of adversarial samples, we chose the ACA (Anti-codec Attack) method for detection. We conducted experiments across datasets of different resolutions. The experimental results are shown in Table X.

From the Table X, frame deletion and frame rate conversion have minimal impact on our method. These operations do not severely affect the CDU pattern division during video encoding, and the mismatch characteristic between the CDU maps and the pixel domain of adversarial samples does not disappear.

In scenarios of video compression with lower compression rates, such as when the QP is set from 20 to 25, the damage to the original content is relatively low, thus allowing for effective detection. However, when the QP is increased to 35, it can eliminate some of the adversarial noise, resulting in a noticeable decrease in the detection rate of adversarial samples. For instance, the detection AUROC score may drop by 0.05 on the HMDB-51 dataset. However, this phenomenon improves with increased resolution, decreasing only 0.0233 in the high-resolution Kinetics-400 dataset. This result is because the CDU patterns in high-resolution datasets are richer, facilitating detection.

Video transcoding can impact detection performance. In this scenario, adversarial videos are first encoded into MPEG4 or H264 formats and then re-encoded using an HEVC encoder

with a QP of 20 for feature extraction. Since MPEG4 and H264 encodings employ different coding operations from HEVC, such as the depth of CDU partitioning and inter-frame and intra-frame prediction modes, some features may be lost, leading to decreased detection effectiveness.

In social media scenarios, we selected uploading to Weibo and transferring videos via WeChat as typical examples. During social media video transmission, video compression and re-encoding are inevitable steps. Both Weibo and WeChat use the H264 encoding method for videos. After downloading the videos from social media, we re-encode them using an HEVC encoder with a QP of 20 for feature extraction before detection. It can be observed that the detection performance significantly degrades after the video has been transmitted, with the maximum decrease reaching 0.1023. However, this phenomenon is mitigated in high-resolution videos, with the maximum decrease being 0.0672. Social media video transmission remains a challenging scenario for adversarial detection.

G. Ablation Study

In this section, ablation experiments will be conducted using the method proposed in this paper to verify the effectiveness of each module within our approach. We will validate our method from two dimensions: feature selection and model architecture. All experiments are tested on the UCF-101 dataset using adversarial sample data produced by the I3D target network. The parameters for the adversarial attack are consistent with the previous experiments.

1) *Compression Domain Feature*: In our method, three CDU maps are simultaneously used. In the ablation experiment, we explored the impact of these three maps on the detection effect. The experimental results are shown in Table XI. It can be seen that CU, PU, and TU all perform well in detecting global attacks. Among them, the performance of CU is slightly lower than the other two. The division of CU is the highest-level division, and the capture of internal texture details and motion information is not as fine as PU and TU, resulting in slightly insufficient representation ability. However, when detecting sparse attacks, CU and TU are difficult to effectively represent, while PU performs better. In sparse attacks, the temporal continuity between frames with perturbations and those without perturbations will be destroyed, resulting in significant changes in inter-frame prediction modes, which in turn affects the division of PU. However, at the CU and TU levels, more attention is paid to changes in spatial texture complexity. Since the number of frames with added perturbations is relatively small, it is difficult to capture the changes in CU and TU.

Using three types of maps for detection can achieve the best detection effect. This approach not only ensures effective representation of motion information features in the time dimension but also allows for feature modeling of finer textures in spatial regions, improving the utilization rate of compressed domain features.

2) *Model Structure*: In the method proposed in this paper, four types of network structures are involved: AP, f_{RGB} , f_{Com} , and the feature fusion classification module. The feature fusion

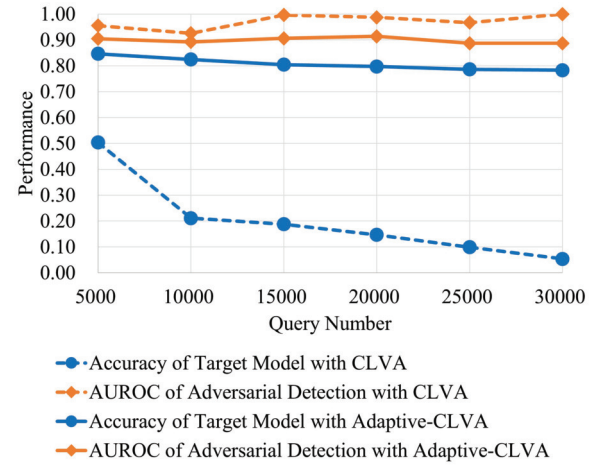


Fig. 11. Experimental results of adaptive attacks, where the dashed line represents ordinary attack and defense scenarios, while the solid line represents adaptive attack scenarios. In an adaptive attack scenario, attackers can query the adversarial detector.

classification module is an essential structure, while AP cannot exist independently without f_{RGB} . Therefore, we have designed four scenarios for discussion: AP+ f_{RGB} , f_{Com} , $f_{RGB} + f_{Com}$, and AP+ $f_{RGB} + f_{Com}$. The experimental results are shown in Table XII. When the input segment contains only pixel domain or compression domain features, the feature fusion classification module will duplicate the input into two copies and feed them separately to ensure the consistency of the output.

The table shows the following conclusions: First, the standalone pixel domain RGB features cannot effectively discriminate against adversarial samples, with the AUROC score for AP+ f_{RGB} being around 0.5. The differences between adversarial and clean samples in the pixel domain are minimal, and the adversarial noise is also very subtle relative to the pixels' value, leading to the network's inability to extract features from adversarial samples effectively. Second, the f_{Com} alone can more effectively detect adversarial samples; for example, the AUROC score for detecting CLVA reaches 0.8997, and for ACA, it reaches 0.8697. This result is because there are more significant statistical differences between adversarial and clean samples in the encoding domain, which can be used for feature extraction and classification by neural networks.

Further, using f_{RGB} in conjunction with f_{Com} achieves better results, with an increase of 0.0562 in the detection of PGD attacks and an increase of 0.1525 in CMA attacks compared to using f_{Com} alone. This result indicates that the fusion of features from different domains can effectively enhance the detection capability for adversarial samples. Additionally, including AP significantly improves detection performance; for instance, it increased by 0.0992 in PGD detection and by 0.1049 in TTA detection. Pre-processing the input frames with AP can effectively reduce the impact of adversarial noise and improve detection accuracy.

In summary, the three modules used in our method, AP, f_{RGB} , and f_{Com} , effectively enhance the ability to detect adversarial video samples.

TABLE XI
COMPRESSION DOMAIN FEATURE ABLATION EXPERIMENTS

CDU maps			Global Attack					Sparse Attack				Average
CU	PU	TU	FGSM	PGD	CMA	TTA	ACA	SA	SVA	AstFocus	CLVA	
✓			0.9042	0.8577	0.9125	0.8778	0.8446	0.6349	0.5032	0.6155	0.6554	0.7562
	✓		0.9446	0.9213	0.9660	0.9553	0.9453	0.9849	0.8454	0.9464	0.9658	0.9417
		✓	0.9263	0.9367	0.8972	0.9112	0.9668	0.6042	0.5225	0.6034	0.6893	0.7842
✓	✓	✓	1.0000	1.0000	0.9950	0.9923	1.0000	0.9999	0.8580	0.9542	0.9778	0.9752

TABLE XII
MODEL STRUCTURE ABLATION EXPERIMENTS

Model Structure			Global Attack					Sparse Attack				Average
AP	f_{RGB}	f_{Com}	FGSM	PGD	CMA	TTA	ACA	SA	SVA	AstFocus	CLVA	
✓	✓		0.5042	0.5563	0.4996	0.5587	0.5223	0.5987	0.5000	0.5000	0.6081	0.5387
	✓	✓	0.8963	0.9008	0.9527	0.8874	0.8845	0.8235	0.7054	0.8394	0.9116	0.8668
		✓	0.8542	0.8446	0.8002	0.7759	0.8697	0.8005	0.6554	0.8235	0.8997	0.8137
✓	✓	✓	1.0000	1.0000	0.9950	0.9923	1.0000	0.9999	0.8580	0.9542	0.9778	0.9752

H. Adaptive Attack Defense

The basic assumption of adaptive attack is that attackers can obtain gradient and structural information of the target model and adversarial detection methods. Specific to the adversarial detection method proposed in this paper, due to the use of compressed domain features during the detection process, and the quantization operation in the video codec process is non-differentiable, so white-box attacks cannot be implemented in our detection network. Therefore, considering the problem of adaptive defense in black-box attacks. That is, attackers can only obtain the output labels and confidence of the target model, as well as the output of the adversarial detector.

$$V_{adv}^{i+1} = CLVA(CLVA(V_{adv}^i, f), AD) \quad (22)$$

The iterative attack step of Adaptive-CLVA is shown in Eq. (22). In i_{th} epoch, the basic CLVA attack $CLVA(\cdot)$ is first used to attack the target model f . Then, the generated adversarial samples V_{adv}^i are used to query the adversarial detector AD and obtain the labels and confidence the adversarial detector gives. The adversarial samples are re-optimized based on the labels and confidence. The attack is successful when the target model outputs the wrong label, and the adversarial detector outputs a clean sample.

The experimental results are shown in Fig. 11. It can be seen that after the introduction of adaptive attacks, the adversarial detection effect proposed by our method has slightly decreased. After 30,000 queries, the AUROC is 0.8872, a decrease of 0.1127, indicating that adaptive attacks in a black box state do have some impact on our method. However, the accuracy of the target model has reached 0.7830, an increase of 0.7294, indicating that our adversarial detection method is a robust adversarial defense method.

VI. CONCLUSION

This paper analyzes that adding adversarial noise to videos will lead to changes in features in the video coding

compression domain, which inspires us to utilize features in the compression domain to detect video adversarial examples. This paper proposes a video adversarial detection method that combines CDU maps features with RGB features. By comparing the consistency between the CDU maps in the compression domain and the decoded RGB image, the adversarial perturbations can be distinguished. Extensive experiments show that our proposed method achieves SOTA detection performance in video adversarial attacks, especially sparse video adversarial attacks, and its generalization is also superior to previous methods.

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